UNIVERSITY OF INFORMATION TECHNOLOGY
FACULTY OF INFORMATION SYSTEMS

CODE 317

COMPUTER TEST
Semester 2 – Academic Year 2024–2025
Subject: MSIS4013 - Database system design, management and administration
Duration: 90 minutes

(***Students are not allowed to use any reference materials***)

Given the database schema “Medical Examination Management” as described in the attribute table below:

TABLE	ATTRIBUTE	DESCRIPTION	DATA TYPE
PATIENT (Stores patient information)	PatientID	Patient identifier	char(5)
	PatientName	Full name of the patient	nvarchar2(50)
	DateOfBirth	Date of birth	date
	NationalID	Citizen identification number	nvarchar2(12)
	HealthInsuranceNo	Health insurance number	nvarchar2(15)
	InsuranceCoverage	Percentage of cost covered by health insurance	float
	Address	Residential address of the patient	nvarchar2(100)
DOCTOR (Stores doctor information)	DoctorID	Doctor identifier	char(5)
	DoctorName	Full name of the doctor	nvarchar2(50)
	StartDate	Employment start date	date
	Specialty	Medical specialty	nvarchar2(50)
MEDICAL_EXAM (Stores patient's examination information)	ExamID	Medical exam identifier	char(5)
	PatientID	Patient identifier	char(5)
	DoctorID	Doctor identifier	char(5)
	ExamDate	Date of examination	date
	Symptoms	Patient's symptoms	nvarchar2(255)
	Diagnosis	Doctor's conclusion	nvarchar2(255)
	FollowUpInDays	Days until next examination	number(5)
MEDICINE (Stores medicine information)	MedicineID	Medicine identifier	char(5)
	MedicineName	Name of the medicine	nvarchar2(50)
	MedicineType	Type of medicine	nvarchar2(50)
	Unit	Unit of measurement (pill, box, bottle)	nvarchar2(20)
	UnitPrice	Unit price (VND)	number(15,2)
PRESCRIPTION (Stores prescription information per exam)	PrescriptionID	Prescription identifier	char(5)
	ExamID	Medical exam identifier	char(5)
	TotalValue	Total value of the prescription (before insurance)	number(15,2)
	InsuranceCoverage	Percentage of cost covered by health insurance	float
	PrescriptionDate	Date the medicine was issued	date
	PatientPayment	Total payment made by patient (after insurance)	number(15,2)
	PrescriptionStatus	Prescription status (Unpaid, Paid)	nvarchar2(30)
PRESCRIPTION_DETAIL (Stores prescribed medicine details)	PrescriptionID	Prescription identifier	char(5)
	MedicineID	Medicine identifier	char(5)
	Quantity	Quantity of medicine provided	number(5)
	Amount	Total cost of the medicine	number(15,2)

Data: https://drive.google.com/file/d/1WWG4ZpgKLZB7AUmBuU_NlcbXMIwJ6hN5/view

Students perform the following tasks:

1. Write the following query using SQL language: List the information of doctors (Doctor ID, Doctor Name) who specialize in "Otolaryngology", along with the information of the patients (Patient ID, Patient Name) they examined in 2024. [1 mark]
2. Write a procedure in PL/SQL that performs the following:
   - Accepts a patient ID as an input parameter
   - Retrieves and displays the following information for that patient: Patient ID, Patient Name, National ID, Address, Number of medical examinations of that patient. Each piece of information should be displayed on a separate line.
   - If the patient ID does not exist, display the message: "Patient with the given ID was not found."

   Write a PL/SQL block to enter a patient ID and invoke this procedure. [2 marks]
3. Write a procedure in PL/SQL that Performs the Following
   - Accept a doctor ID as input.
   - If the doctor does not exist, display the message: "Doctor with the given ID was not found."
   - If the doctor does exist, for each examination, display the following information: Exam ID, Exam Date, Patient Name, Symptoms, Diagnosis. Each piece of information should be displayed on a separate line.
   - If the doctor has never performed any medical examinations, display the message: "No medical examinations found for this doctor."

   Write a PL/SQL block to enter a doctor ID and invoke the procedure. [2.5 marks]
4. Write a function in PL/SQL that performs the following:
   - Accepts a patient ID as input. The function returns a value of type NUMBER.
   - Returns the total amount of money the patient has paid for all prescriptions (associated with the patient's medical examinations).
   - If the patient does not exist, return -1.
   - If the patient exists but has no prescriptions or has not paid for any, return 0.

   Write a PL/SQL block to enter a patient ID, and display the following information: Patient ID, Total amount paid by the patient (retrieved by calling the function above) [2.5 marks]
5. Create a trigger to enforce the following constraint: "A prescription (PrescriptionID) must not include more than 10 different types of medicine (MedicineID)." [2 marks]

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